Exam 3 — Synthesis and transfer

Questions only

Name: _____ **Date:** _____

Conditions

- 50 questions: 30 single-answer multiple choice and 20 numerical. Each is worth 1 raw practice point.
- Suggested training limit: **150 minutes**. This is not a claim about the actual exam duration or question count.
- Use up to three double-sided A4 sheets, prepared by you, and a local Jupyter notebook. No internet, AI assistance or answer key during the attempt.
- Use Python/NumPy for computation; SciPy is also allowed. You do not need executable OpenCV or PyTorch. Specialized routines are provided when needed.
- For multiple choice, select exactly one of A–D. Attempt every question.
- Unless an exact integer is requested, round the final numerical answer to **two decimal places**. Keep full precision in intermediate steps. Use fractions rather than percentages unless specified.
- Numerical practice grading uses the syllabus rule: within ± 0.01 of the unrounded reference, inclusive. Counts require the exact integer; no partial credit.
- Conventions stated in a question override defaults. Arrays use [row,column] = [y,x]; geometric vectors are columns; image-plane projection is virtual/front-facing; foreground is 1.
- This is original, challenging training material based on the supplied course files, not a lecturer-issued mock or a prediction of the exam.

Solutions are in a separate PDF. Do not open them until you finish.

Part A — Single-answer multiple choice

Q01 — Multiple choice

A unit normal n_c is expressed in the camera frame. The camera uses $X_c = RX_w + t$. Which expression converts the normal to the world frame?

A. $n_w = R^\top (n_c - t)$

B. $n_w = Kn_c$, followed by perspective division

C. $n_w = R^\top n_c$; translation does not act on a direction vector.

D. $n_w = Rn_c + t$

Q02 — Multiple choice

You resize an image to one quarter of each spatial dimension by keeping every fourth pixel. Why should a suitable low-pass filter normally be applied BEFORE subsampling?

A. Keeping every fourth pixel is itself an ideal low-pass filter, so preprocessing cannot influence aliasing.

B. I_t guarantees exact recovery of every original sample, even when the image has frequencies above the coarse-grid limit.

C. High spatial frequencies can otherwise alias into false lower-frequency patterns.

D. Low-pass filtering after subsampling removes aliasing just as reliably as filtering before subsampling.

Q03 — Multiple choice

A single image is modelled as $I = A \odot S$. Without further constraints, can it uniquely determine both albedo and shading?

A. No, but dividing every pixel by 255 always resolves the ambiguity.

B. No; multiple albedo/shading combinations can produce the same image.

C. Yes, because one RGB pixel always supplies all geometry and material parameters.

D. Yes, if the image has a high enough resolution, regardless of assumptions.

Q04 — Multiple choice

Why can directly fitting the linear photometric-stereo model to strongly gamma-compressed pixel values be wrong?

A. Normalizing the fitted g removes all gamma effects, so only albedo can be affected by a nonlinear response.

B. The model assumes a linear relation between measured intensity and radiance; the response must be linearized or modelled.

C. Gamma can always be absorbed into one constant camera gain, independent of intensity and light direction.

D. No correction is needed: any common gamma exponent preserves the linear equations exactly under a change of albedo alone.

Q05 — Multiple choice

What determines a sensor channel's response to an illuminated coloured surface?

A. The interaction of source spectrum, surface spectral reflectance and that channel's spectral sensitivity.

B. Only the surface's colour name, independent of light and sensor.

C. Only the maximum channel value in a different pixel.

D. Only the camera's extrinsic translation.

Q06 — Multiple choice

An integrable normal-derived gradient field is reconstructed on a connected image domain with known x/y spacing and at least one nonzero slope. Which ambiguity remains without an anchor or boundary height?

A. No height ambiguity remains, because unit-normal normalization already fixes the absolute reference height.

B. An additive constant in height.

C. An arbitrary multiplicative height scale, even though p and q and the x/y spacing are fixed.

D. An independent additive height constant for every pixel, even while all neighbouring derivatives stay fixed.

Q07 — Multiple choice

Apply a normalized averaging kernel to a constant positive image with zero padding. What is correct?

A. All outputs become zero because averaging removes the DC component.

B. Every output must be identical, including boundaries, for any padding rule.

C. Only corners become brighter because there are fewer neighbours.

D. Interior outputs remain constant, but outputs near boundaries can be darker.

Q08 — Multiple choice

A closing operation uses a small structuring element. Which statement is safest?

A. I_t can fill sufficiently small gaps/holes relative to the element's shape, but need not fill a large cavity.

B. Closing removes every isolated foreground component without changing holes.

C. Closing means erosion followed by dilation.

D. One closing always fills every hole in every binary object.

Q09 — Multiple choice

Erode an all-ones finite image using a centred 3 by 3 all-ones element with outside-image values defined as zero. What happens?

A. Its foreground expands by one pixel outside the image.

B. I_t remains all ones because every stored pixel is one.

C. Its outermost pixel border is removed.

D. Every interior pixel becomes zero while the border stays one.

Q10 — Multiple choice

Circularly translate a discrete image by an integer offset. How does the DFT transform under the standard shift theorem?

A. Only magnitude changes; phase is fixed.

B. The spectrum is only reordered by fftshift; individual frequency coefficients never change.

C. Both magnitude and phase must stay unchanged because the image content is the same.

D. Each coefficient is multiplied by a frequency-dependent phase factor of unit magnitude; its magnitude is unchanged.

Q11 — Multiple choice

For a real-valued image, what relationship holds between opposite-frequency DFT coefficients?

A. Their imaginary components must all be zero, because the original image contains only real values.

B. They are complex conjugates, with indices interpreted modulo the image size.

C. They are always identical in both real and imaginary components, so their phases must also be identical.

D. They are always negatives of each other, so the spectrum is antisymmetric at every opposite-frequency pair.

Q12 — Multiple choice

A detected intensity edge does not necessarily mark a depth discontinuity. Why?

A. Canny hysteresis guarantees that only physical depth boundaries survive its high threshold.

B. A material boundary cannot affect intensity, so every strong edge must be a depth discontinuity.

C. Increasing Gaussian σ makes intensity edges uniquely attributable to changes in depth.

D. Reflectance or illumination changes can also cause an intensity edge.

Q13 — Multiple choice

Why does SIFT search across a Gaussian scale space and use Difference-of-Gaussians extrema?

A. To locate distinctive structures at their characteristic scales; DoG approximates a scale-normalized LoG response.

B. To recover absolute object size in metres from one image.

C. To guarantee invariance to arbitrary projective deformation of every patch.

D. To force all keypoints to have identical pixel scale.

Q14 — Multiple choice

For positive affine brightness changes $I' = aI + b$ with $a > 0$, why can a normalized gradient-based SIFT descriptor be relatively robust?

A. Dividing pixel coordinates by the focal length removes b.

B. Gradients remove the constant offset b, and descriptor normalization reduces the common gain a.

C. The descriptor stores the original unnormalized RGB values exactly.

D. SIFT estimates and removes every spatially varying shadow without assumptions.

Q15 — Multiple choice

An affine RANSAC sample contains three correct correspondences, but the three source points are collinear. Why should the sample be rejected?

A. Three pairs supply six scalar equations, so the transform is necessarily unique despite collinearity.

B. I_t is geometrically degenerate and does not uniquely constrain a general 2D affine transform.

C. An all-inlier sample is always sufficient; degeneracy can occur only when at least one correspondence is wrong.

D. Repeated least-squares refitting of the same three pairs automatically restores the missing rank.

Q16 — Multiple choice

Increasing the RANSAC inlier threshold gives a model many more counted inliers. Why is this not automatically evidence of a better geometric model?

A. A larger threshold always improves localization accuracy.

B. A threshold has no influence on consensus size.

C. The larger threshold can admit poor matches and outliers as inliers.

D. Every counted inlier has exactly zero residual by definition.

Q17 — Multiple choice

A stationary scene undergoes a moving cast shadow. A brightness-constancy flow method reports motion in the shadow region. Which interpretation is correct?

A. The appearance-constancy assumption for a physical surface point is violated; apparent image motion need not equal physical surface motion.

B. Increasing the window size always establishes that reported image motion corresponds to physical surface motion.

C. The reported vector must be a sign error, because a static scene guarantees zero estimated flow regardless of illumination.

D. Every measured optical flow vector necessarily equals a 3D object's velocity.

Q18 — Multiple choice

Why can a coarse-to-fine flow method still fail on a tiny fast-moving object?

A. Small objects always have zero displacement at full resolution.

B. The coarse-level flow already gives exact depth in metres.

C. Pyramids make every flow observable regardless of texture or visibility.

D. The object can disappear or become indistinct at coarse levels, so its motion is not initialized correctly.

Q19 — Multiple choice

An online tracker updates its appearance model from its own predicted box. What is a major risk during occlusion?

A. Any online update guarantees improved test accuracy.

B. Incorrect predictions can contaminate the model and cause tracking drift.

C. Occlusion automatically provides perfect new positive labels.

D. The model can never change because the initial template is fixed by definition.

Q20 — Multiple choice

A camera translates while observing a scene with nearby and distant objects. One fitted panorama homography aligns the distant background but not the foreground. What is the main cause?

A. Depth-dependent parallax; one global homography need not align the nonplanar scene.

B. The foreground and background require opposite matrix multiplication orders, even when the transform direction is fixed.

C. The blend ramp is too narrow; widening it guarantees correction of every geometric mismatch.

D. H has the wrong homogeneous scale; multiplying all its entries by a different nonzero scalar corrects foreground without moving background.

Q21 — Multiple choice

H maps source image coordinates to a reference image. A translation T then moves reference coordinates into a larger panorama canvas. With column vectors, which matrix maps source directly to the canvas?

A. TH

B. HT

C. $H + T$

D. $H^{\mathsf{T}}T$, regardless of the coordinate convention

Q22 — Multiple choice

Which classifier and prediction does this code implement for binary class labels? Squared distances preserve Euclidean neighbour order. Provided routine: `np.argsort(d2)` returns indices sorted by increasing d2.

```
import numpy as np
X = np.array([[0,0], [0,1], [1,0], [3,3]], dtype=float)
y = np.array([1,0,0,1])
query = np.array([.1,.1])
d2 = ((X-query)**2).sum(axis=1)
nearest = np.argsort(d2)[:3]
prediction = int(y[nearest].sum() > 1)
```

A. K-means predicts class 0 after moving three learned centres; the array y is used only as an initialization seed.

B. Unweighted 3-nearest-neighbour voting predicts class 1 because the closest example's label outweighs both others.

C. Unweighted 3-nearest-neighbour majority voting predicts class 0, although the closest example has class 1.

D. 1-nearest-neighbour classification predicts class 1; the two other selected labels never contribute.

Q23 — Multiple choice

For pixel clustering, why might features $[R, G, B, \alpha x, \alpha y]$ give different segments from $[R, G, B]$?

A. K-means always returns connected regions even with colour-only features.

B. Adding x,y guarantees the true semantic labels without supervision.

C. Spatial terms encourage location coherence, and α sets their relative importance against colour differences.

D. Feature scaling cannot affect Euclidean distances.

Q24 — Multiple choice

A training set contains label noise and is not linearly separable. Why use a soft-margin rather than a hard-margin linear SVM?

A. I_t removes every mislabeled point with certainty.

B. I_t guarantees infinite margin for nonseparable data.

C. I_t permits margin violations while balancing fit against regularization.

D. I_t ignores the weight norm entirely and only memorizes pixels.

Q25 — Multiple choice

A CNN outputs softmax class scores that sum to one. Which conclusion is NOT guaranteed merely by applying softmax?

A. Increasing one logit relative to the others increases its softmax probability.

B. Each output is nonnegative.

C. The outputs sum to one, up to numerical precision.

D. The probabilities are well calibrated to actual correctness frequencies.

Q26 — Multiple choice

Consider this training sequence from the kind of loop taught in Lab5. What is the purpose of the first line? No executable PyTorch code is required.

```
optimizer.zero_grad()
loss.backward()
optimizer.step()
```

A. Disable gradient recording for this batch while retaining earlier batches' stored gradients.

B. Reset learned model parameters to zero, ensuring that each batch starts from the same initialized network.

C. Normalize every stored gradient to unit norm, so the learning rate becomes irrelevant to the next update.

D. Prevent unintended accumulation of earlier batches' gradients before this batch's update.

Q27 — Multiple choice

A residual branch changes feature shape from $H \times W \times C$ to $\frac{H}{2} \times \frac{W}{2} \times 2C$. What is required for an additive shortcut?

A. A compatible shortcut, such as a stride-2 projection, so the tensors have matching shapes.

B. Remove all nonlinearities from the entire network.

C. Add the mismatched tensors with no mapping; addition always changes the channel count automatically.

D. Concatenate them and call the result an identity addition.

Q28 — Multiple choice

Multiply both a predicted box and its ground-truth box coordinates by the same positive scale. Which property is correct?

A. Both losses must be identical for every pair of boxes.

B. IoU doubles whenever object area doubles.

C. Coordinate error becomes scale invariant just by summing its four components.

D. IoU is unchanged, while an unnormalized squared coordinate-error loss generally scales by the square of that factor.

Q29 — Multiple choice

For an ideal rectified stereo pair with parallel optical axes and a horizontal baseline, why are epipolar lines parallel?

A. The epipoles are at infinity in the horizontal baseline direction.

B. The fundamental matrix maps every point to the image's principal point.

C. Both cameras have exactly the same optical centre.

D. Every scene point has the same depth.

Q30 — Multiple choice

Which description of a standard multi-view reconstruction pipeline is correct?

A. Bundle adjustment modifies only colour histograms and never camera parameters.

B. SfM estimates camera poses and sparse structure; bundle adjustment refines poses and points jointly; MVS can densify using estimated cameras.

C. SfM requires only one camera image with no priors or known geometry.

D. MVS always estimates metric scale from arbitrary uncalibrated image content alone.

Part B — Numerical values

Q31 — Numerical

An original camera has $f_x = 800$, $f_y = 600$, $c_x = 320$, $c_y = 240$ and zero skew. Crop away 40 pixels from the left and 20 from the top; then halve both spatial dimensions. Use continuous coordinates with no half-pixel offsets. A camera-frame point is $\begin{bmatrix}0.5 & 1 & 2\end{bmatrix}$. Report its final image v coordinate, using the virtual image plane in front of the camera.

Answer: _____

Q32 — Numerical

For the RGB image below, compute normalized RGB PER PIXEL, then report the spatial mean of its normalized red channel. For black pixels, define all normalized channels as zero.

```
import numpy as np
I = np.array([[[6,1,3], [2,6,2]],
              [[0,0,0], [1,1,2]]], dtype=float)
s = I.sum(axis=2, keepdims=True)
safe = np.where(s == 0, 1, s)
normalized = I / safe
answer = normalized[:, :, 0].mean()
```

Provided function: `np.where(condition, a, b)` selects a where the condition is true and b elsewhere. Black pixels are included in the spatial mean.

Answer: _____

Q33 — Numerical

Let estimated height gradients be $p(x,y) = \frac{\partial z}{\partial x} = y$ and $q(x,y) = \frac{\partial z}{\partial y} = 2x$. Start with $z(0,0) = 0$. Integrate continuously to (2,3) along path A: x first then y; and path B: y first then x. Report the absolute difference between the two resulting heights.

Answer: _____

Q34 — Numerical

Use calibrated Lambertian intensity $I = \rho \max(n \cdot l, 0)$, with unit vectors $n = \begin{bmatrix}2/3 & -1/3 & 2/3\end{bmatrix}^\mathsf{T}$ and $l = \begin{bmatrix}0 & 0.6 & 0.8\end{bmatrix}^\mathsf{T}$, $\rho = 0.75$, and unit light strength/gain. Report I.

Answer: _____

Q35 — Numerical

A 9 by 9 binary image is zero except for a 3 by 3 all-ones block at rows 3..5 and columns 3..5 (inclusive, zero-based), and a single foreground pixel at (0,0). Apply opening with a centred 3 by 3 all-ones element, zero outside the image, retaining the same shape after each stage. Report the final foreground-pixel count.

Answer: _____

Q36 — Numerical

Use the median of three entries: `med(v)=sorted(v)[1]`. Let $a = \begin{bmatrix}0 & 0 & 10\end{bmatrix}$ and $b = \begin{bmatrix}0 & 10 & 0\end{bmatrix}$, with elementwise addition. Report $\text{med}(a+b) - \text{med}(a) - \text{med}(b)$ as an exact integer.

Answer: _____

Q37 — Numerical

For $f = \begin{bmatrix}1 & 0 & 2 & 0\end{bmatrix}$ and $h = \begin{bmatrix}0 & 1 & 0 & 1\end{bmatrix}$, use 4-point CIRCULAR convolution $c_k = \sum_{n=0}^{3} f_n h_{(k-n) \bmod 4}$. Report c_3 as an exact integer. This is the result of multiplying their unpadded 4-point DFTs and applying the inverse DFT.

Answer: _____

Q38 — Numerical

The grid below contains gradient magnitudes AFTER non-maximum suppression. Use 8-neighbour Canny hysteresis: strong means ≥ 9; weak means ≥ 4 and < 9; lower values are rejected. Retain every weak pixel connected through weak/strong pixels to a strong pixel. Report the retained pixel count.

$$\begin{bmatrix}10 & 6 & 0 & 0 & 5 & 5 & 0\\ 0 & 6 & 0 & 0 & 0 & 5 & 0\\ 0 & 6 & 6 & 0 & 0 & 0 & 0\end{bmatrix}$$

Answer: _____

Q39 — Numerical

Use equal weights for a four-pixel window with $I_x = \begin{bmatrix}1 & 0 & 1 & 2\end{bmatrix}$ and $I_y = \begin{bmatrix}0 & 2 & 1 & 0\end{bmatrix}$. Construct

$$M = \begin{bmatrix}\sum I_x^2 & \sum I_x I_y\\ \sum I_x I_y & \sum I_y^2\end{bmatrix}.$$

Report the Shi–Tomasi score, the smaller eigenvalue. Provided formula for symmetric

$$\begin{bmatrix}a & b\\ b & d\end{bmatrix}:$$

$\lambda_{\min} = \frac{a+d-\sqrt{(a-d)^2+4b^2}}{2}$.

Answer: _____

Q40 — Numerical

A flow at one-eighth spatial resolution is $[u, v] = \begin{bmatrix}1.75 & -0.5\end{bmatrix}$ in COARSE pixels. Upsample to full resolution with the correct vector-unit scaling, then add a full-resolution residual flow $\begin{bmatrix}0.25 & 0.75\end{bmatrix}$. Report the Euclidean magnitude of the resulting full-resolution vector.

Answer: _____

Q41 — Numerical

An affine RANSAC uses minimal samples of 3 point pairs and independently sampled inlier fraction $w = 0.4$. Assume all-inlier samples are nondegenerate. Use $N = \left\lceil \frac{\log(1-p)}{\log(1-w^3)} \right\rceil$. Report the minimum integer N for $p = 0.999$.

Answer: _____

Q42 — Numerical

A nonsingular homography maps these four source points to targets:

$$(0,0) \mapsto (1,2)$$
$$(1,0) \mapsto (3,2)$$
$$(0,1) \mapsto (1,4)$$
$$(1,1) \mapsto (3,3)$$

Use

$$H = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & 1 \end{bmatrix},$$

so $x' = \frac{ax+by+c}{gx+hy+1}$, $y' = \frac{dx+ey+f}{gx+hy+1}$. Rearrange the eight scalar equations into a linear system. You may use the provided least-squares routine `np.linalg.lstsq(A, rhs, rcond=None)[0]` to solve a full-rank system. Report the target x coordinate for source (0.5,0.5).

Answer: _____

Q43 — Numerical

A matcher returns SQUARED Euclidean distances $[D_1, D_2]$, nearest first. Apply Lowe's test to the ordinary distances: $\frac{\sqrt{D_1}}{\sqrt{D_2}} < 0.7$. For the rows below, report the largest ordinary nearest distance among accepted matches.

$$\begin{bmatrix} 4 & 9 \\ 9 & 16 \\ 1 & 4 \\ 25 & 36 \\ 0.25 & 1 \end{bmatrix}$$

Answer: _____

Q44 — Numerical

A 1D visual vocabulary has centres $\begin{bmatrix} 0 & 4 & 8 \end{bmatrix}$. Quantize descriptors $\begin{bmatrix} 1 & 3 & 5 & 7 & 8 & 9 \end{bmatrix}$ to their nearest centre using Euclidean distance. Build counts, then L1-normalize by the number of descriptors. Report the normalized bin value for centre 8.

Answer: _____

Q45 — Numerical

A binary classifier predicts positive iff score ≥ 0.6. True labels are $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$ and scores are $\begin{bmatrix} 0.9 & 0.8 & 0.7 & 0.6 & 0.5 & 0.4 \end{bmatrix}$, in the same order. Report F1 for the positive class as a fraction.

Answer: _____

Q46 — Numerical

One class has 4 ground-truth objects. After score-ordered, one-to-one matching, ranked detections are [FP,TP,FP,TP,TP]. Use ALL-POINTS INTERPOLATED AP: sum each positive recall increment times the maximum precision at that recall or any later recall. Precision beyond maximum achieved recall is zero. Report AP as a fraction. This is NOT 11-point AP or raw uninterpolated AP.

Answer: _____

Q47 — Numerical

A residual block has an identity shortcut and this dense bottleneck branch: 1×1 convolution $64 \to 16$ channels; 3×3 convolution $16 \to 16$; 1×1 convolution $16 \to 64$. Include one bias per output channel in EACH convolution. There is no normalization layer or trainable shortcut. Report total trainable parameters.

Answer: _____

Q48 — Numerical

A standard 3D convolution processes a video tensor with 2 input channels and uses 5 output channels. Each filter has temporal x height x width size $3 \times 3 \times 3$. Include one bias per output channel. No groups. Report the trainable parameter count; the video length and spatial image size do not enter this count.

Answer: _____

Q49 — Numerical

Use the ORIGINAL YOLO v1 confidence convention: box confidence $=$ $\Pr(\text{object}) \cdot \text{IoU}(\text{predicted box}, \text{GT})$; class-specific score $=$ $\Pr(\text{class} \mid \text{object}) \cdot$ box confidence. Given $\Pr(\text{object}) = 0.75$, $\text{IoU} = 0.6$ and $\Pr(\text{class} \mid \text{object}) = 0.8$, report the class-specific score.

Answer: _____

Q50 — Numerical

For rectified stereo, $Z = \frac{fB}{d}$. Let $f = 900$ pixels, $B = 0.2$ metres, $d = 12$ pixels, and small disparity-error magnitude $\delta_d = 0.25$ pixel. Use FIRST-ORDER error propagation $|\delta_Z| \approx \frac{fB}{d^2}|\delta_d|$. Report the estimated depth-error magnitude in metres, not the exact finite-difference error.

Answer: _____